BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Arithmetic, Comparison and Casting

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does assignment do to a variable?

2. What is the CPU's main job?
   A) Execute instructions
   B) Keep permanent file backups
   C) Display every image
   D) Supply electrical power

3. A folder contains 30 images of 200 KB each. Using decimal units, give its total size in KB and MB.

4. A photographer travels with large files and needs fast transfers. "The cheapest storage is always best." Correct this. What else matters?

Today's objectives

Apply arithmetic and comparison operators and type conversion. Use arithmetic operators and comparison operators in a practical scenario.

Key vocabulary - Read, cover, recall

Arithmetic operators

Operators such as +, -, * and / allow programs to calculate results.

Comparison operators

Symbols such as ==, < and >= compare values and produce Boolean results.

Casting

int() converts appropriate text or numeric values to integer so arithmetic can use them.

Order of operations

Parentheses can make intended calculation order clear and avoid ambiguous results.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

What is Arithmetic operators?

Arithmetic operators - Operators such as +, -, * and / allow programs to calculate results.

In this example

A train fare calculator uses comparison operators; Symbols such as ==, < and >= compare values and produce Boolean results.

Follow the effect

int() converts appropriate text or numeric values to integer so arithmetic can use them. Explain how this changes the eventual program result.

Check understanding

Order of operations is used for a specific reason: Parentheses can make intended calculation order clear and avoid ambiguous results.

I Do - Worked model

A train fare calculator is being written. Explain how arithmetic operators and comparison operators are used to make the program work. [4 marks]

- We are designing a train fare calculator. Identify which data and decisions are essential.
- Arithmetic operators [1] - Operators such as +, -, * and / allow programs to calculate results. [1].
- Comparison operators [1] - Symbols such as ==, < and >= compare values and produce Boolean results. [1].
- Two linked points for each technique give four mark points.

We Do - Guided practice

1. In a museum ticket price, explain how arithmetic operators helps the program. [2 marks]

2. For a retail discount, explain why comparison operators matters. [2 marks]

3. When designing a sports points, explain how casting and order of operations should be used. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Convert the text "4" into an integer and print how many bits are in four bytes.

```
bytesText = "4"
bytes = ...
print(bytes * 8)
```

You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A school trip cost needs to work correctly. Explain how comparison operators and casting would help. [4 marks]

Q4. A pupil is programming a takeaway bill. Explain the roles of arithmetic operators, casting and order of operations in their solution. [6 marks]

Exit Ticket - True or False?

Operators such as +, -, * and / allow programs to calculate results.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It stores a value in the named variable, replacing its previous value if one existed.

2. A - Execute instructions

3. 6,000 KB = 6 MB.

4. Check capacity, speed, cost and how it will be used. An SSD can suit fast transfers and travel, if its capacity and price meet the photographer's needs.

I Do - Explain the model

We are designing a train fare calculator. Identify which data and decisions are essential.

Arithmetic operators [1] - Operators such as +, -, * and / allow programs to calculate results. [1].

Comparison operators [1] - Symbols such as ==, < and >= compare values and produce Boolean results. [1].

Two linked points for each technique give four mark points.

We Do - Model answers

1. Arithmetic operators [1] - Operators such as +, -, * and / allow programs to calculate results. [1].

2. Comparison operators [1] - Symbols such as ==, < and >= compare values and produce Boolean results. [1].

3. Casting [1] - int() converts appropriate text or numeric values to integer so arithmetic can use them. [1]. Order of operations [1] - Parentheses can make intended calculation order clear and avoid ambiguous results. [1].

Code First - Answer

```
bytesText = "4"
bytes = int(bytesText)
print(bytes * 8)
```

Expected output: 32

You Do - Self-mark

Q2: Arithmetic operators [1]; Casting [1].

Q3: Comparison operators [1] - Symbols such as ==, < and >= compare values and produce Boolean results. [1]. Casting [1] - int() converts appropriate text or numeric values to integer so arithmetic can use them. [1].

Q4: Arithmetic operators [1] - Operators such as +, -, * and / allow programs to calculate results. [1]. Casting [1] - int() converts appropriate text or numeric values to integer so arithmetic can use them. [1]. Order of operations [1] - Parentheses can make intended calculation order clear and avoid ambiguous results. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.